



Attachment 1: Questions to Accompany the Bottled Liquid Scanner Request for Information

General Systems Description:

1. Describe the fundamental operating principles of the bottled liquid scanner?
 - a. What core technologies are employed in the system's operation?
 - b. Provide block diagrams/schematic illustrating major system components and functionality.
2. What algorithm does the system use for material discrimination and identification?
3. What are the dimensions of the system?
4. What is the weight of the system?
5. What is the system's operating wavelength (if applicable)?
6. What is the system's source power output, and is it adjustable?

User Dependency & Safety Considerations:

7. What safety considerations are involved in system operation?
 - a. What personal protective equipment (PPE) is required for operators?
 - b. Can the system be modified for safe operation without requiring PPE in a checkpoint screening environment?
8. What level of experience or technical knowledge is required for system users?
 - a. What general training is required for Transportation Security Officers (TSOs) to operate the system?
 - b. Are any additional qualifications necessary for operators to achieve proficiency?
 - c. Does the system require input from an operator (e.g., Transportation Security Officers) to assist in detection or identification?
9. Can the system be adapted to support multiple user modes (e.g., a basic mode for TSOs and an advanced mode for Transportation Security Specialists – Explosives (TSS-E))?



Material Detection & Identification Capabilities:

10. Can the system differentiate between threat and non-threat materials?
11. Is the system capable of performing chemical identification?
12. What specific materials/substances can the system effectively detect based on TSA's prohibited items list? The list can be found here: <https://www.tsa.gov/travel/security-screening/whatcanibring/all>
13. Does the system include a built-in material/substance database?
 - a. What materials and how many substances are included in the database? Provide a list if possible.
 - b. Does the database include common consumer goods such as food, beverages, toiletries, baby products, over-the-counter medicines, and pharmaceuticals?
 - c. Can users update the database to include new benign substances for improved performance?
 - d. How does the size of the material database affect system analysis time?
14. Can the system optimize analysis time by restricting the library to common items relevant to airport security screening?

Container Compatibility:

15. Can the system analyze liquid samples within their original containers without opening or removing a sample?
16. What types of container materials can the system scan through (e.g., transparent, translucent, opaque)?
17. What are the physical parameters, or size limitations of compatible containers?
18. Can the system scan through:
 - a. Clear, colored, and opaque glass containers?
 - b. Frosted glass containers?
 - c. All colors of glass?
 - d. Foil-lined containers?



- e. Metal pouches?
 - f. Plastic storage bags and rigid plastic containers?
 - g. Security Tamper-Evident Bags (STEBs)?
 - h. Other materials such as ceramics, wood, or composites?
19. How does the system handle scanning through printed, painted, or labeled container surfaces?
20. Does the system notify users when printed materials or labels obstruct an effective scan?

Multi-Component & Mixture Analysis:

21. What capabilities does the system have for analyzing multi-component mixtures?
22. What is the maximum number of components the system can differentiate in a non-monomolecular mixture?
23. What factors impact the system's ability to accurately analyze and differentiate components in multi-component mixtures?

Sample Handling & Analysis:

24. What is the minimum and maximum sample size required for analysis?
25. What is the focal distance of the scanning system, and can it be adjusted?
26. What physical states can the system analyze (solids, liquids, gases)?
27. Can the system effectively analyze powders, gels, emulsions, and granular solids?
28. Can the system differentiate between white, light-colored, and dark-colored powders?
29. How well does the system handle loosely packed or "fluffy" materials?
30. Is the system sensitive to density variations or gradients in material packing?

Performance & Speed:

31. What is the total analysis time from sample introduction to results output in seconds?
32. How does analysis time vary based on container material (e.g., plastic vs. glass vs. metal) and liquid type?
33. Can the system perform real-time analysis, or is there a delay between scanning and result output?



34. Does the system provide any audio or visual resolution indicators? If so, describe. Is it adjustable?
35. What is the mean time between scans (i.e., time required before scanning the next sample)?
36. What is the maximum throughput (i.e., how many containers can be scanned per minute)?
37. Can the system be optimized to reduce scanning time for high-traffic airport checkpoints?
38. What factors contribute to extended analysis times, and how can they be mitigated?
39. Does reducing the material/substance database size impact scan speed?

False Alarm Rates & System Accuracy:

40. What is the system's false alarm rate when analyzing common benign liquids (e.g., water, juice, baby formula, medications)?
41. What is the system's nuisance alarm rate (i.e., alarms triggered by non-threatening substances that require secondary screening)?
42. What factors could contribute to false alarms, nuisance alarms, inconclusive results, or extended analysis times?
43. What is the average resolution time for a false alarm (i.e., how long does it take an operator to determine it is non-threatening)?
44. How does the false alarm rate vary across different container materials (e.g., glass, plastic, metal) and liquid compositions?
45. What measures are in place to reduce false positives while maintaining security effectiveness?
46. What is the percentage of "inconclusive" results, and how long does it typically take to resolve them?
47. What environmental or operational factors contribute to false alarms or extended analysis times?
48. Can the system notify users to rescan a container if the scan was obstructed by printed materials or labels, and what is the added delay?
49. What is the system's detection confidence level (e.g., probability of correctly identifying a threat liquid vs. a non-threat liquid)?



- 50. How does the system handle borderline cases, and what percentage of scans require operator intervention for further analysis?
- 51. If a false alarm occurs, how long does it take for the system to retest the same sample and provide a new result? Is this process automated?

System Adaptability & Optimization:

- 52. Can the system's scanning speed be adjusted based on the required level of accuracy and detection confidence?
- 53. What are the trade-offs between faster scan times and detection accuracy?
- 54. How does the system prioritize speed vs. accuracy, and can this be configured by security personnel?
- 55. How quickly can the system recalibrate or reset if an error or anomaly occurs?

Regulatory Compliance & Certification:

- 56. What other industries or organizations currently utilize this equipment? Provide independent third-party or government test reports, if available.
- 57. Provide performance data demonstrating the system's effectiveness in checkpoint alarm resolution.
- 58. Has the system been submitted for European Civil Aviation Conference (ECAC) evaluation, and if so, what were the results?
 - a. If ECAC-certified, what is the detection performance level (e.g., Type A, B, C) assigned to the system?
 - b. Provide any ECAC-recognized independent test reports available to verify system efficacy.
 - c. Are there any modifications required to meet ECAC certification standards?
- 59. If the system is not ECAC-approved, what steps are being taken toward compliance, and what is the expected timeline?